

1) Scores on the Wechsler Adult Intelligence Scale, a standard IQ test, are approximately Normal for the age group of 20 to 34. The mean score is 110 with a standard deviation of 25. What percent of this age group has an IQ below 100?

a) Write the inequality using the raw score.

b) Find the z-score and rewrite the inequality.

c) Draw the Normal standardized curve.

d) What are you entering into your calculator?

Lower:

Upper:

Mean (μ):

Standard Deviation (σ):

e) _____% of adults from 20 to 34 have an IQ less than 100.

2) Suppose the systolic blood pressure of an adult male is normally distributed with a mean of 138 mm of mercury and standard deviation of 10.

If an adult male is picked at random, find the percentage of men whose systolic blood pressure will be

(a) greater than 160 mm

(b) between 120 and 135 mm

(c) less than 110 mm

(d) between 130 and 140 mm

Find the systolic blood pressure for men that represents:

(e) the 90th percentile

(f) the bottom 4%

3) A special task force of a military unit requires that the recruits not be too tall or too short. Suppose 12% of the applicants are rejected because they are too tall and 18% because they are too short. If the height of an applicant is normally distributed with a mean of 69.4 inches and a standard deviation of 3.5 inches, determine the heights that define whether an applicant is accepted or rejected.

4) If $z = 1.5$, $\bar{x} = 22$, and $s = 8.5$, what is your "x" value?

5) If $z = -0.55$, $\bar{x} = 13.4$, and $s = 2$, what is your "x" value?

6) If you have 20% of your data lying below a z-score, what is that z-score?

7) If you have 34.5% of your data lying above a z-score, what is that z-score?

8) If you have 75% of the data lying symmetrically in the middle of your distribution, what are your z-scores?

9) What is the z-score that corresponds to the first quartile? Third quartile?

10) A college professor plans to give a test and to curve the results. The raw test scores are normally distributed with a mean of 58 and a standard deviation of 9. She wants the grades to be divided up as follows:

D	C	B	A
Bottom 25%	50 th percentile or lower before a D	Between an A and a C	Top 10%

What will be the raw score averages she gives to each of the letter grades?